

SHANGHAI BAOSHAN, China

WMO: 583620

Lat: 31.3908N

Lon: 121.4447E

Elev: 30

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.6	30.8	10.3	9.3	35.7	14.7	11.6	36.6	17.2	39.3	15.8	39.4	5.7	270	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	95.9	80.0	93.6	79.7	91.4	79.2	82.1	90.5	81.2	89.3	80.5	88.2	7.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.0	155.7	86.6	79.1	151.3	85.8	78.4	147.6	85.3	46.0	90.9	45.1	89.7	44.3	88.5	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	14.5	13.1	24.6	98.9	2.5	2.2	22.8	100.5	21.3	101.8	19.9	103.0	18.1	104.6
			21.6	83.7	2.9	1.3	19.6	84.6	17.9	85.4	16.3	86.1	14.2	87.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	41.1	43.7	50.9	60.7	69.9	76.1	84.5	83.8	76.9	67.8	57.1	45.8	(6)
(7)		DBStd	15.90	5.79	6.22	6.54	6.35	4.92	4.80	4.91	4.10	4.44	4.82	6.57	6.36	(7)
(8)		HDD50	720	283	193	67	3	0	0	0	0	0	15	159		(8)
(9)		HDD65	2815	740	596	440	159	11	0	0	0	27	246	596		(9)
(10)		CDD50	5577	7	17	94	324	618	783	1070	1049	807	550	229	29	(10)
(11)		CDD65	2197	0	0	2	30	164	333	605	584	357	112	10	0	(11)
(12)		CDH74	21091	0	1	17	191	866	2356	7679	7100	2592	272	17	0	(12)
(13)		CDH80	8505	0	0	2	40	216	729	3783	3084	622	29	0	0	(13)
(14)	Wind	WSAvg	6.7	6.4	6.7	7.0	7.0	6.9	6.5	6.9	7.5	7.0	6.3	6.2	6.4	(14)
(15)	Precipitation	PrecAvg	45.2	2.0	2.3	3.3	3.6	4.1	6.5	5.2	6.7	5.4	2.4	2.3	1.7	(15)
(16)		PrecMax	65.8	7.7	5.6	6.8	7.6	9.5	19.1	14.2	17.9	13.9	11.7	4.7	6.1	(16)
(17)		PrecMin	29.8	0.0	0.3	0.5	1.2	0.9	1.5	0.9	0.0	0.5	0.0	0.1	0.0	(17)
(18)		PrecStd	8.7	1.4	1.4	1.5	1.6	1.9	3.3	3.1	4.4	3.4	2.2	1.5	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	67.6	75.9	85.0	90.0	95.0	99.3	98.5	93.0	83.5	76.3	65.1	(19)
(20)		0.4%	MCWB	55.0	58.2	60.6	67.9	68.1	78.7	79.7	80.2	78.7	72.8	66.2	58.1	(20)
(21)		2%	DB	56.2	61.6	70.0	79.3	85.5	90.8	96.8	95.1	89.0	79.3	72.1	61.0	(21)
(22)		2%	MCWB	50.3	53.9	57.6	64.7	67.9	77.8	79.7	80.1	78.0	69.6	64.4	54.7	(22)
(23)		5%	DB	52.5	56.7	65.6	74.9	81.9	87.3	94.6	92.6	85.7	76.8	69.2	57.7	(23)
(24)		5%	MCWB	47.8	50.3	55.5	63.7	67.8	76.3	80.0	79.7	76.9	68.0	62.4	52.6	(24)
(25)		10%	DB	49.4	52.7	61.5	71.1	78.8	84.0	92.4	90.4	83.3	74.7	66.4	55.0	(25)
(26)		10%	MCWB	45.4	47.6	53.7	61.0	66.7	75.0	79.8	79.5	75.7	66.5	60.7	50.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.0	60.3	63.0	71.6	75.9	81.0	83.7	82.9	81.8	76.1	70.1	61.4	(27)
(28)		0.4%	MCDWB	61.1	65.9	70.0	79.0	83.8	90.3	92.9	91.8	89.0	81.1	74.0	63.4	(28)
(29)		2%	WB	52.0	55.0	60.2	68.5	73.4	79.5	82.4	81.8	79.9	72.9	66.8	56.4	(29)
(30)		2%	MCDWB	54.5	59.5	66.6	75.1	79.1	87.9	91.2	89.9	85.7	76.9	70.1	59.5	(30)
(31)		5%	WB	48.8	51.4	57.6	66.0	71.5	78.2	81.6	81.0	78.4	70.6	64.0	53.7	(31)
(32)		5%	MCDWB	51.4	54.9	63.5	72.0	76.6	85.3	90.1	88.5	83.9	74.4	67.5	56.9	(32)
(33)		10%	WB	45.8	48.5	54.9	63.3	69.6	76.7	80.8	80.4	77.1	68.6	61.8	51.0	(33)
(34)		10%	MCDWB	48.9	52.0	60.5	69.2	74.7	82.3	89.1	87.4	82.4	73.2	65.2	54.2	(34)
(35)	Mean Daily Temperature Range		MDBR	9.7	10.1	12.0	12.9	12.3	9.6	10.4	9.2	8.7	9.7	10.8	10.7	(35)
(36)		5% DB	MCDBR	15.5	18.1	19.9	19.7	18.2	14.9	13.9	12.7	11.2	12.0	13.9	14.6	(36)
(37)		5% DB	MCWBR	10.0	10.9	10.2	9.8	7.5	5.6	4.3	4.1	4.4	5.3	7.3	9.3	(37)
(38)		5% WB	MCDBR	11.7	14.6	16.5	15.6	12.3	12.2	12.1	10.6	9.8	8.9	10.4	10.6	(38)
(39)		5% WB	MCWBR	9.5	10.6	10.8	9.8	7.6	5.4	4.8	4.3	4.6	5.9	7.5	9.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.590	0.631	0.664	0.671	0.657	0.715	0.689	0.654	0.602	0.603	0.614	0.584		(40)
(41)		taud	1.677	1.624	1.587	1.629	1.694	1.627	1.713	1.768	1.844	1.758	1.688	1.702		(41)
(42)		Ebn at Noon	190	199	206	213	217	204	209	214	219	204	182	182		(42)
(43)		Edh at Noon	64	74	82	82	77	82	75	70	63	64	63	60		(43)
(44)	All-Sky Solar Radiation	RadAvg	721	841	1198	1408	1518	1325	1665	1548	1265	1076	796	705		(44)
(45)		RadStd	91	161	100	161	164	194	200	209	124	143	95	96		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	-0.90	N/A	N/A	N/A	+122	
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	N/A	(47)

Nomenclature: See separate page